

Chukwuemeka K. Onuoha

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EDUCATION

Bachelor of Science in Electrical Engineering, Minor in Engineering Mathematics | GPA: 3.38
Saint Louis University

May 2027
St. Louis, MO

RELATED EXPERIENCE

Undergraduate Researcher

January 2026 – Present

Saint Louis University | John Allen

St. Louis, MO

- Developing an Android application to evaluate adaptive audio-haptic feedback for visually impaired users.
- Implementing vibration modulation via Android haptic APIs to evaluate perceptual thresholds.
- Designing calibration protocols for detectable and comfortable vibration limits.
- Integrating text-to-speech and multimodal cues while applying spatial cognition research to guide system architecture and experimental design.

Electrical Engineering Intern

May 2025 – August 2025

Crawford, Murphy & Tilly (CMT)

St. Louis, MO

- Developed electrical schematics and layouts for multiple design projects.
- Conducted NEC-compliant electrical load and sizing calculations.
- Engineered panels, transformers, lighting, and HVAC load designs using AutoCAD, AGI32, Visual, and Revit.
- Verified construction progress through site visits and collaboration with senior engineers

PROJECTS

Vowel Classification Using Spectral Band Energy Features | *MATLAB*

Nov. 2025

- Built a speech classification system using Fourier-domain spectral band energy and nearest-neighbor rules.
- Evaluated classifier performance with confusion matrices and analyzed feature separability.

Amplitude Modulation & Demodulation with RMSE Analysis | *MATLAB*

Oct. 2025

- Implemented AM modulation/demodulation of multiple audio signals with custom MATLAB functions.
- Analyzed sensitivity to carrier-frequency offsets, observing RMSE growth with increasing offset.

Echo Cancellation Using Convolution | *MATLAB*

Sept. 2025

- Designed an audio echo removal system using autocorrelation-based delay estimation and inverse filtering.
- Developed a calibration curve for echo gain mapping via simulated trials and polynomial fitting.

SKILLS

- Tools: AutoCAD, Revit, AGI 32, Visual, Multisim, Questa, Vivado, Waveforms.
- Programming: Python, MATLAB, Kotlin (working knowledge).

RELEVANT COURSEWORK

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|-----------------------------|---------------------|------------------|-------------------|
| • Electronic Circuit Design | • Energy Conversion | • Semiconductors | • Thermodynamics |
| • Automatic Control Systems | • Microprocessors | • Linear Systems | • Digital Design. |

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

- President, SLU Ping Pong Club: Led club operations, events, recruitment, and executive board.
- Residential Advisor: Supported residents and first-year students through mentoring and community engagement.
- Recreational Center Specialist: Ensured safe, fair, and positive recreational environments.
- Calling Agent: Engaged alumni to strengthen relationships and support fundraising initiatives.

PROFESSIONAL SOCIETY

- Event Coordinator, Institute for Electrical and Electronics Engineers (IEEE) – Student Branch
- Member, National Society of Black Engineers (NSBE)